POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

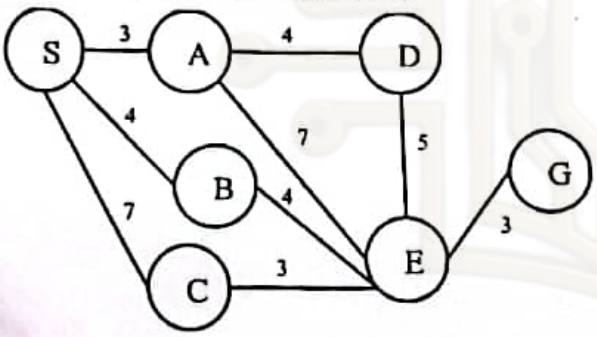
Attempt all the questions.

a) Define Artificial Intelligence. Is the ChatGPT an intelligent agent? If yes, explain in detail. 7

b) What is PEAS? Explain the architecture of reflex and goal-based agents with an example such as a vacuum cleaning agent. 8

a) Design an intelligent agent that can solve a sliding tile puzzle (8-puzzle). The agent can slide tiles in different directions to reach the goal configuration. Formulate/model this problem as a production system. 8

b) What is A* search? Apply A* search to determine the optimal path from Node S to Node G. 7



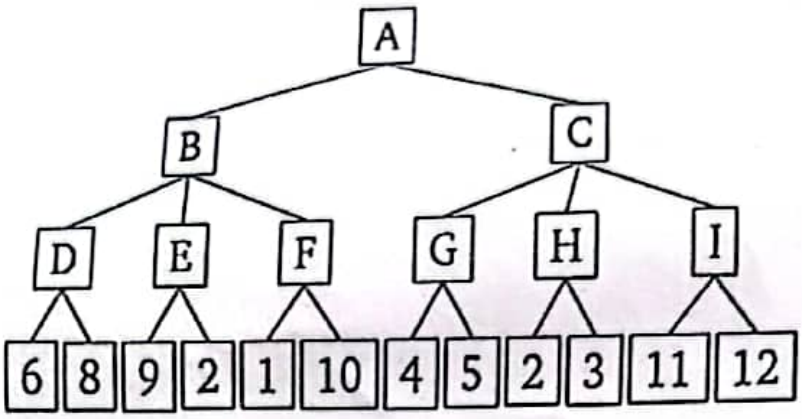
NODES	HEURISTIC
S	7
A	6
B	4
C	5
D	2
E	2
G	0

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3. a) What is the idea of Mini-Max search algorithm? Explain how the alpha-beta pruning works with reference to the following game tree. Assume the root node is MAX node. 8



b) What is Knowledge Representation? Discuss how Predicate logic extends the expressive power of propositional logic in detail. 7

a) Define machine learning. Explain the learning by analogy approach with suitable examples. 7

b) What is Clustering? Explain how the K-Means Clustering Algorithm works with an example. 8

a) What is support vector machine? Explain in detail. 7

b) What is Artificial Neural Network? Explain the perceptron training algorithm. Highlight the limitation of the perceptron training algorithm and mention a scenario where it might fail to perform well. 8

a) What is an expert system? Explain the architecture of an expert system in detail. 8

b) Provide examples of ambiguity at the phonetic, syntactic, semantic, and pragmatic levels in Natural Language Processing. 7

Write short notes on: (Any two) 2×5

a) Turing Test

b) Fuzzy Logic

c) Reasoning under uncertainty

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